

and FDA regulation on Electronic Records and Electronic Signatures.⁴ Electronic structures and standards related to electronic submissions are out of scope for this guidance.

FDA's guidance documents, including this guidance, do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidances means that something is suggested or recommended, but not required.

II. BACKGROUND

With the use of computerized systems for capturing clinical investigation data, it is common to find at least some source data recorded electronically. Common examples include, but are not limited to clinical data initially recorded in electronic health records maintained by healthcare providers and institutions, electronic laboratory reports, digital medical images from devices, and electronic diaries completed by study subjects.

FDA regulations define an *electronic record* as any combination of text, graphics, data, audio, pictorial, or other information represented in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system.⁵ An eCRF is an example of an electronic record.

The eCRF is an auditable electronic record of information that generally is reported to the sponsor on each trial subject, according to a clinical investigation protocol. The eCRF enables clinical investigation data to be systematically captured, reviewed, managed, stored, analyzed, and reported.

Source data includes all information in original records and certified copies of original records of clinical findings, observations, or other activities in a clinical investigation used for reconstructing and evaluating the investigation.⁶ Access to source data is critical to the review and inspections of clinical investigations. The review of source data by both the FDA and sponsor is important to ensure adequate protection of the rights, welfare, and safety of human subjects and the quality and integrity of the clinical investigation data. Source data should be attributable, legible, contemporaneous, original, and accurate (ALCOA) and must meet the regulatory requirements for recordkeeping.⁷ Capturing source data electronically and transmitting it to the eCRF should:

⁴ See 21 CFR part 11.

⁵ 21 CFR 11.3(b)(6).

⁶ In 21 CFR 312.62(b), reference is made to records that are part of case histories as “supporting data”; the ICH guidance for industry *E6 Good Clinical Practice: Consolidated Guidance* (the ICH E6 guidance) (available at <http://www.fda.gov/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/default.htm> or <http://www.fda.gov/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/default.htm>) uses the term “source data/documents.” For the purpose of this guidance, these terms describe the same information and have been used interchangeably.

⁷ For the principal record keeping requirements for clinical investigators and sponsors developing drugs and biologics, see 21 CFR 312.50, 312.58, 312.62, and 312.68. For medical devices, see 21 CFR 812.140 and 812.145.

- Eliminate unnecessary duplication of data
- Reduce the possibility for transcription errors
- Encourage entering source data during a subject's visit, where appropriate
- Eliminate transcription of source data prior to entry into an eCRF
- Facilitate remote monitoring of data
- Promote real-time access for data review
- Facilitate the collection of accurate and complete data

III. ELECTRONIC SOURCE DATA

Electronic source data are data initially recorded in electronic format. They can include information in original records and certified copies of original records of clinical findings, observations, or other activities captured prior to or during a clinical investigation used for reconstructing and evaluating the investigation.

A. Data Capture

1. Electronic Source Data Origination

A *data element* in an eCRF represents the smallest unit of observation captured for a subject in a clinical investigation. Examples of data elements include race, white blood cell count, pain severity measurement, or other clinical observations made and documented during a study. Each data element is associated with an authorized data originator. Examples of data originators include, but are not limited to:

- Clinical investigator(s) and delegated clinical study staff
- Clinical investigation subjects or their legally authorized representatives
- Consulting services (e.g., a radiologist reporting on a computed tomography (CT) scan)
- Medical devices (e.g., electrocardiograph (ECG) machine and other medical instruments such as a blood pressure machine)
- Electronic health records (EHRs)
- Automated laboratory reporting systems (e.g., from central laboratories)
- Other technology

A list of all authorized data originators (i.e., persons, systems, devices, and instruments) should be developed and maintained by the sponsor and made available at each clinical site. In the case of electronic, patient-reported outcome measures, the subject (e.g., unique subject identifier) should be listed as the originator.

When identification of data originators relies on identification (log-on) codes and unique passwords, controls must be employed to ensure the security and integrity of the authorized user

names and passwords.⁸ When electronic thumbprints or other biometric identifiers are used in place of an electronic log-on/password, controls should be designed to ensure that they cannot be used by anyone other than their original owner.

When a system, device, or instrument automatically populates a data element field in the eCRF, a data element identifier (see section III.A.3) should be created that automatically identifies the particular system, device, or instrument (e.g., name and type) as the originator of the data element. For example, if an ECG machine automatically transmits to the eCRF, a data element identifier should be generated that identifies the ECG machine as the originator.

2. *Source Data Capture*

Data can be entered into the eCRF either manually or electronically as described below.

a. *Direct Entry of Data Into the eCRF*

Many data elements (e.g., blood pressure, weight, temperature, pill count, resolution of a symptom or sign) in a clinical investigation can be obtained at a study visit and can be entered directly into the eCRF by an authorized data originator. This direct entry of data can eliminate errors by not using a paper transcription step before entry into the eCRF. For these data elements, the eCRF is the source. If a paper transcription step is used, then the paper documentation should be retained and made available for FDA inspection (see section III.A.2.c).

When pertinent supportive information is available, FDA could request other documents during an inspection to corroborate a direct entry of source data elements into the eCRF by an authorized data originator. For example, at an initial visit, a clinical investigator might ask a subject about underlying illnesses, and proceed to enter the illness(es) in an eCRF. During an FDA inspection, a record may be requested for evidence of testing or the use of medications to corroborate a diagnosis.

Typically, images (e.g., CT scans) are not included as data elements in an eCRF, but rather the clinical interpretation of the image is included as a predefined data field. When an image (e.g., a CT scan) is sent to a central reading center for clinical interpretation and the radiologist is authorized to enter data directly into the eCRF, then the radiologist's assessment (e.g., *normal*) is the data element as predefined in the eCRF, the radiologist is the data originator and the CT scan is the pertinent clinical record. However, when the radiologist sends a report to a clinical investigator(s) who transcribes the data into the eCRF, the clinical investigator(s) is the originator and the source is the radiologist's report.

b. *Automatic Transmission of Data From Devices or Instruments Directly to the eCRF*

When a device or instrument is the data originator (e.g., blood pressure monitoring device or glucometer) and data are automatically transmitted directly to the eCRF, the eCRF is the source. If a process is used by which the device or instrument transmits data automatically to, for

⁸21 CFR 11.300(a); see also the computerized systems guidance.